



COMPETENCY STANDARD

FOR

TIG AND MIG WELDING

(CONSTRUCTION SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for TIG and MIG Welding (Construction sector) is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

This document has been developed to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)** to meet the skills requirements of the Construction sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Occupation Specific Competencies (100 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-CON-TMW-01-O	Apply occupational health and safety (OHS) practice in the workplace	1. Apply personal health and safety practices. 2. Report hazards and risks in welding practice 3. Respond to emergencies.	04
SICIP-CON-TMW-02-O	Carry Out Shielded Metal Arc Welding	1. Identify and prepare work requirements 2. Select welding job, equipment and job holding devices 3. Perform welding 4. Clean/maintain the workplace	08
SICIP-CON-TMW-03-O	Perform Gas Welding	1. Prepare for gas welding 2. Perform gas welding 3. Perform gas cutting 4. Clean and store tools and equipment	08
SICIP-CON-TMW-04-O	Carry out Gas Metal Arc Welding	1. Identify and prepare work requirements 2. Select welding job, equipment and job holding devices 3. Perform GMAW or MIG welding 4. Clean/maintain the workplace	40
SICIP-CON-TMW-05-O	Carry out Gas Tungsten Arc Welding	1. Identify and prepare work requirements 2. Select welding job, equipment and job holding devices 3. Perform GTAW or TIG welding 4. Clean/maintain the workplace	40
Total Hours			0100

COMPETENCY STANDARD: TIG and MIG Welding

The Occupation Specific Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 04 Hrs.	Unit Code: SICIP-CON-TMW-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practice in the workplace. It specifically includes the task of applying personal health and safety practices, reporting hazards and risks in welding practice and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply personal health and safety practices	1.1 <u>Personal protective equipment (PPE)</u> are identified and explained. 1.2 Personal Protective Equipment (PPE) is worn according to the task requirements. 1.3 The workspace is kept clean and organized to reduce the risk of accidents. 1.4 Proper hand hygiene practices, such as washing hands or using hand sanitizer, are followed regularly to prevent contamination. 1.5 Safety protocols are adhered to, including the proper handling and disposal of hazardous materials and waste.
2. Report hazards and risks in welding practice	2.1 Potential welding <u>hazards and risks</u> are identified and reported immediately. 2.2 Accurate information is communicated to supervisors or safety officers. 2.3 Welding equipment is inspected regularly to avoid hazards and risks. 2.4 Any defects or malfunctions in welding machine are reported for immediate repair. 2.5 Ventilation systems are monitored to ensure proper airflow.
3. Respond to emergencies	3.1 Alarming and warning devices are identified. 3.2 <u>Emergency response plans and procedures</u> are responded to. 3.3 <u>First aid procedures</u> during emergency situations are identified.

Range of Variables

Variable	Range (Includes but not limited to):
1. Personal protective equipment	1.1 Safety glasses 1.2 Ear plugs 1.3 Gloves 1.4 Apron 1.5 Helmet 1.6 Mask 1.7 Safety shoes
2. Hazards and risks	2.1 Open flames 2.2 Exposure to fumes 2.3 Electrical hazards 2.4 Hot surfaces
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify and explain the types of Personal Protective Equipment (PPE) based on the task and their uses. 1.2 Keep the workspace clean and organized to minimize the risk of accidents. 1.3 Regularly practice proper hand hygiene to prevent contamination. 1.4 Follow safety protocols, ensuring the correct handling and disposal of hazardous materials and waste. 1.5 Recognize welding hazards. 1.6 Regularly inspect welding equipment. 1.7 Ensure proper ventilation system. 1.8 Recognize the first aid procedures in emergency situations.
2. Underpinning Skills	2.1 Wearing the correct PPE as required for specific welding tasks. 2.2 Maintaining a clean and organized workspace to minimize the risk of accidents. 2.3 Conducting regular inspections of welding equipment to identify and avoid risks. 2.4 Reporting any defects or malfunctions in welding machines promptly for repairs.

	2.5 Responding promptly to emergency response plans and procedures to ensure safety. 2.6 Applying first aid procedures during emergency situations to provide immediate assistance.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 applied personal health and safety practices 1.2 reported hazards and risks in welding practice 1.3 responded to emergencies
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT SHIELDED METAL ARC WELDING	Nominal Duration: 08 Hrs.	Unit Code: SICIP-CON-TMW-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out shielded metal arc welding. It specifically includes the task of identifying and preparing work requirements, selecting welding job, equipment and job holding devices, performing welding and cleaning/maintaining the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and prepare work requirements	1.1 Welding plans are developed based on the specifications provided. 1.2 Safe work practices are observed and PPE are worn. 1.3 Drawings are interpreted to shielded metal arc welding conforming to the specifications. 1.4 <u>Welding machine</u> and <u>Tools and equipment</u> are selected according to the requirements. 1.5 <u>Base metal/plates</u> and <u>electrodes</u> are selected according to requirements of the job. 1.6 Base metals are prepared as per requirement. 1.7 <u>Welding joint and position</u> are demonstrated according to the job requirement.
2. Select welding job, equipment and job holding devices	2.1 Welding equipment and holding devices are set up and adjusted in accordance with the job requirements. 2.2 Welding area guards, work table/floor, dust collection devices are checked according to worksite procedure. 2.3 Welding machine is prepared and bead practices in different positions are carried out.
3. Perform welding	3.1 Welding machine performance is checked conforming to the job requirement. 3.2 Welding in <u>different positions</u> are carried out. 3.3 Welds are cleaned, checked for quality and <u>defects</u> are identified. 3.4 Defects are rectified to meet the standards of job specifications.
4. Clean/maintain the workplace	4.1 Workplace is inspected to identify any potential hazards. 4.2 Cleaning tasks are performed according to the cleanliness protocols. 4.3 Work areas are organized, and tools and equipment are stored. 4.4 Equipment and machinery are cleaned and maintained. 4.5 The workplace is kept free from debris and obstructions to ensure a safe and efficient working environment.

Range of Variables

Variable	Range (Includes but not limited to):
1. Welding Machine	1.1 AC Arc welding machine (dry/wet). 1.2 DC Arc welding machine.
2. Tools and equipment	2.1 C-clamp 2.2 Chipping hammer 2.3 Ball pin hammer 2.4 Hack saw 2.5 File 2.6 Chisel 2.7 Center punch 2.8 Blacksmith tong 2.9 Pliers 2.10 Wire brush 2.11 Weld gauge 2.12 Welding table with positioner 2.13 Job holding devices/fixture 2.14 Hand grinder 2.15 Hand drill
3. Base metal plates	3.1 MS Plates 3 mm to 12 mm thickness
4. Electrodes	4.1 8,10 and 12 SWG (4, 3.2 and 2.5 mm)
5. Welding joint and position	5.1 F-means fillet. 5.2 G-means groove 5.3 These two welds are 4 (four) in position. 5.4 Pos. No-1 is flat position. 5.5 Pos. No-2 is Horizontal position. 5.6 Pos. No-3 is Vertical position. 5.7 Pos. No-4 is Overhead position. 5.8 Example-1F indicates the flat position of fillet welding 5.9 1G indicates the flat position of groove welding and so on.
6. Different positions	6.1 Butt joint 6.2 Lap joint 6.3 Tee joint 6.4 Corner joint 6.5 Edge joint
7. Defects	7.1 Lack of penetration 7.2 Excess of penetration 7.3 Porosity 7.4 Crack 7.5 Spatter 7.6 Slag Inclusion 7.7 Distortion 7.8 Undercut 7.9 Lack of fusion 7.10 Notch

	7.11 Irregular shape 7.12 Dimension
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Interpret welding drawings for Shielded Metal Arc Welding (SMAW) 1.2 Assess the type of materials and thickness for selecting the right welding method. 1.3 Understand the importance of securing the workpiece to ensure safety and accuracy during welding. 1.4 Understand the importance of controlling the welding environment to prevent contamination. 1.5 Importance of proper cleaning to prevent defects and ensure the durability of the weld. 1.6 Understand the keeping the equipment in good working condition. 1.7 Understand the importance of maintaining a clean work environment.
2. Underpinning Skills	2.1 Developing welding plans based on job specifications. 2.2 Selecting and preparing the appropriate welding machine, tools, and equipment. 2.3 Selecting base metals, plates, and electrodes according to the welding job's requirements. 2.4 Adjusting welding equipment and job holding devices. 2.5 Preparing the welding machine and perform bead practice in various positions. 2.6 Performing welding in different positions. 2.7 Cleaning welds and inspect for quality. 2.8 Performing cleaning tasks according to cleanliness protocols to maintain a safe environment. 2.9 Cleaning and maintaining equipment and machinery to keep them in good working condition.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Welding machine 4.4 Base plates 4.5 Electrodes 4.6 Tools and equipment 4.7 Materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and prepared work requirements 1.2 selected welding job, equipment and job holding devices 1.3 performed welding 1.4 cleaned/maintained the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM GAS WELDING	Nominal Duration: 08 Hrs.	Unit Code: SICIP-CON-TMW-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform gas welding. It specifically includes the task of preparing for gas welding, performing gas welding, performing gas cutting and cleaning and storing tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for gas welding	1.1 PPE is selected & used. 1.2 <u>Tools, welding equipment</u> and <u>materials</u> are selected and collected. 1.3 Gas welding sets and accessories are gathered and set up according to the job specifications. 1.4 Drawings are interpreted to produce component to the job specifications. 1.5 Jig, fixture and positioner are set up for gas welding.
2. Perform gas welding	2.1 Gas welding torch is adjusted to perform different types of flame for welding. 2.2 Gas welding set performance is checked conforming to the job requirement. 2.3 Gas welding is performed in different <u>welding joint and position</u> 2.4 Welds are cleaned, checked for quality and defects are identified. 2.5 Defects are rectified to meet the standards of job specifications.
3. Perform gas cutting	3.1 Gas cutting torch is set for cutting materials as per job thickness. 3.2 Gas cutting set performance is checked conforming to the job requirement. 3.3 <u>Gas cutting</u> is performed as per the job requirement. 3.4 Cut pieces rough edges are removed, cleaned, checked for quality. 3.5 Defects are identified and corrective action is taken according to standard cutting procedures.
4. Clean and store tools and equipment	4.1 Tools, equipment and cutting torch are cleaned. 4.2 Work place is cleaned 4.3 Waste materials are disposed in proper place. 4.4 Tools, equipment and finished job are stored safely in appropriate location according to standard place and procedures.

Range of Variables

Variable	Range (Includes but not limited to):
1. Tools	1.1 C-clamp 1.2 Chipping hammer 1.3 Ball pin hammer 1.4 Center punch 1.5 Hack saw 1.6 File 1.7 Chisel 1.8 Pliers 1.9 Wire brush 1.10 Weld gauge 1.11 Blacksmith tong 1.12 Welding table with positioner 1.13 Job holding devices/fixture 1.14 Hand grinder 1.15 Hand drill 1.16 Soldering tools
2. Welding equipment	2.1 Gas welding set and its accessories 2.2 Oxygen cylinder 2.3 Acetylene cylinder 2.4 Cylinder pressure gauge (oxy/acetylene) 2.5 Working pressure gauge (oxy/acetylene) 2.6 Oxygen regulator 2.7 Oxygen hose 2.8 Spark lighter 2.9 Apparatus wrench 2.10 Welding torch 2.11 Torch tip/ nozzle 2.12 Oxygen hose connections 2.13 Acetylene hose connections 2.14 Acetylene regulator 2.15 Acetylene hose 2.16 Twin hose 2.17 Welder hose joints 2.18 Coupler joint socket 2.19 Coupler joint plug 2.20 Cutting torch
3. Materials	3.1 Mild steel 3.2 Carbon steel 3.3 Cast iron 3.4 Aluminum 3.5 Brass 3.6 Brazing materials 3.7 Soldering materials
4. Welding joint and	4.1 Joints – Lap, butt, Tee and corner.

position	4.2 Position –Flat and Horizontal
5. Gas cutting	5.1 Straight cutting 5.2 Zig zag cutting 5.3 L cutting 5.4 Round cutting

Curricular Content Guide

1. Underpinning Knowledge	1.1 Understand the importance of selecting the correct personal protective equipment (PPE). 1.2 Recognize the process of gathering and setting up the gas welding set. 1.3 Importance of setting up jigs, fixtures, and positioners. 1.4 Recognize the need to adjust the gas welding torch. 1.5 Importance of checking the performance of the gas welding set. 1.6 Procedure for performing gas welding in various joints and positions. 1.7 Cleaning process of welds and how to check for defects or quality issues. 1.8 Process of checking the gas cutting set. 1.9 Process of performing gas cutting based on the job requirements. 1.10 Importance of removing rough edges from cut pieces and checking them for quality. 1.11 Recognize the importance of keeping the workplace clean and free from debris.
2. Underpinning Skills	2.1 Selecting and wearing appropriate PPE (Personal Protective Equipment). 2.2 Selecting and collecting the necessary tools, welding equipment, and materials. 2.3 Adjusting the gas welding torch to produce different types of flame. 2.4 Checking the performance of the gas welding set. 2.5 Performing gas welding on various joints and in different positions. 2.6 Cleaning welds and inspect them for quality, identifying any defects or imperfections. 2.7 Adjusting the gas cutting torch to the correct settings based on the material thickness for accurate cutting. 2.8 Performing gas cutting according to the job requirements, ensuring precision and quality. 2.9 Cleaning tools, equipment, and the cutting torch after use.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn

	3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Gas welding machine 4.5 Gas cutting machine 4.6 Materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for gas welding 1.2 performed gas welding 1.3 performed gas cutting 1.4 cleaned and stored tools and equipment
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT GAS METAL ARC WELDING	Nominal Duration: 40 Hrs.	Unit Code: SICIP-CON-TMW-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out gas metal arc welding. It specifically includes the tasks of identifying and preparing work requirements, selecting welding job, equipment and job holding devices, performing GMAW or MIG welding and cleaning/maintaining the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and prepare work requirements	1.1 Welding plans are developed based on the specifications provided. 1.2 Safe work practices are observed and PPE are worn. 1.3 GMAW or MIG welding drawings are interpreted. 1.4 MIG welding machine, <u>Tools and equipment</u> are selected according to the requirements. 1.5 <u>Base metal/ plates, wire electrode size and shielding gas</u> are selected according to requirements of the job. 1.6 Base metals and <u>GMAW weld area</u> are prepared as per requirement. 1.7 MIG welding machine operations is demonstrated 1.8 <u>Welding joint and position</u> are demonstrated according to the job requirement.
2. Select welding job, equipment and job holding devices	2.1 Welding job, equipment and holding devices are identified as per requirement. 2.2 MIG welding machine is prepared as per job requirement. 2.3 Welding equipment and holding devices are set up and adjusted in accordance with the job requirements.
3. Perform GMAW or MIG welding	3.1 MIG welding machine and welding torch performance is checked in accordance with the job requirement. 3.2 Amperage and gas flow is set in accordance with work piece plate thickness. 3.3 Welding is performed in Butt and Tee joints in flat and horizontal positions. 3.4 Welds are cleaned, checked for <u>quality test</u> and <u>defects</u> are identified. 3.5 Defects are rectified to meet the standards of job specifications.
4. Clean/maintain the workplace	4.1 Workplace is inspected to identify any potential hazards. 4.2 Cleaning tasks are performed according to the cleanliness protocols. 4.3 Work areas are organized, and tools and equipment are stored.

	4.4 Equipment and machinery are cleaned and maintained.
	4.5 The workplace is kept free from debris and obstructions to ensure a safe and efficient working environment.

Range of Variables

Variable	Range (Includes but not limited to):
1. Tools and equipment	1.1 C-Clamp 1.2 Chipping hammer 1.3 Ball pin hammer 1.4 File 1.5 Chisel 1.6 Hack saw 1.7 Blacksmith tong 1.8 Scriber 1.9 Pliers 1.10 Wire brush 1.11 Weld gauge 1.12 Welding table with positioner 1.13 Job holding devices/fixture 1.14 Hand grinder 1.15 Hand drill 1.16 MIG torch 1.17 Gas regulators 1.18 MIG wire and consumables 1.19 Torch tips 1.20 Torch neck liner 1.21 Torch neck assembly 1.22 Gas valve 1.23 Torch trigger assembly 1.24 Gas flow tester 1.25 Gas flow meter 1.26 Wire spool holder 1.27 Wire spool assembly 1.28 Wire feed
2. Base metal/ plates	2.1 MS Plate 2.2 Carbon steel 2.3 Stainless steel 2.4 Aluminum
3. Wire electrode size	1.1 Wire diameter- 0.7 – 2.4 mm
4. Shielding gas	4.1 Inert gas- Argon, helium 4.2 Reactive gases – oxygen or carbon dioxide
5. GTAW weld area	5.1 Direction of travel 5.2 Shielding gas 5.3 Contact tube 5.4 Wire electrode (consumable) 5.5 Molten metal

	5.6 Solidified weld metal. 5.7 Work piece. 5.8 Electric arc. 5.9 Copper shoe.
6. Welding joint and position	6.1 Butt joint and Tee joint. 6.2 F, 2F and 1G, 2G
7. Quality test	7.1 Destructive test: 7.1.1 Tensile test 7.1.2 Bend test 7.1.3 Hardness test 7.1.4 Break test 7.2 Non-Destructive test: 7.2.1 Radiographic test 7.2.2 Ultrasonic test 7.2.3 Magnetic particle test 7.2.4 Leak test 7.3 Visual test
8. Defects	8.1 Lack of penetration 8.2 Excess of penetration 8.3 Porosity 8.4 Crack 8.5 Slag Inclusion 8.6 Distortion 8.7 Spatter 8.8 Undercut 8.9 Lack of fusion 8.10 Notches 8.11 Irregular shape 8.12 Dimension 8.13 Poor weld start 8.14 Wire stubbing 8.15 Excessive spatter 8.16 Burn through 8.17 Spatter 8.18 Convex bead

Curricular Content Guide

1. Underpinning Knowledge	1.1 Interpret GMAW (Gas Metal Arc Welding) or MIG (Metal Inert Gas) welding drawings. 1.2 MIG welding machines, tools, and equipment according to the specific requirements of the welding job. 1.3 Select the appropriate base metals, wire electrode size, and shielding gas based on the job's specifications. 1.4 Determine the operation of MIG welding machines. 1.5 Check the performance of the MIG welding machine and welding torch. 1.6 Understand the amperage and gas flow rate based on the thickness of the workpiece plate.
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	1.7 Procedure for performing MIG welding on butt and tee joints in both flat and horizontal positions. 1.8 Understand the importance of cleaning and maintaining welding equipment and machinery.
2. Underpinning Skills	2.1 Developing welding plans based on provided specifications. 2.2 Selecting and collecting MIG welding machines, tools, and equipment. 2.3 Preparing base metals and the GMAW weld area. 2.4 Demonstrating the operation of the MIG welding machine. 2.5 Demonstrating proper welding joints and positions. 2.6 Preparing the MIG welding machine. 2.7 Performing welding on butt and tee joints in flat and horizontal positions. 2.8 Cleaning welds and inspect them for quality. 2.9 Cleaning and maintaining equipment and machinery to ensure their longevity and proper functioning.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 MIG welding machine 4.5 Materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and prepared work requirements 1.2 selected welding job, equipment and job holding devices 1.3 performed GMAW or MIG welding 1.4 cleaned/maintained the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of	3.1 Competency assessment must be done in an

Assessment	assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: CARRY OUT GAS TUNGSTEN ARC WELDING	Nominal Duration: 40 Hrs.	Unit Code: SICIP-CON-TMW-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out gas tungsten arc welding. It specifically includes the tasks of identifying and preparing work requirements, selecting welding job, equipment and job holding devices, performing GTAW or TIG welding and cleaning/maintaining the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and prepare work requirements	1.1 Welding plans are developed based on the specifications provided. 1.2 Safe work practices are observed and PPE are worn 1.3 Gas tungsten arc welding drawings are interpreted. 1.4 TIG welding machine, <u>Tools and equipment</u> are selected according to the requirements. 1.5 <u>Base metals/plate, filler metal</u> , Tungsten <u>electrodes</u> and <u>shielding gas</u> are selected. 1.6 Base metals and <u>GTAW welds area</u> are prepared as per requirement. 1.7 <u>Welding joints and positions</u> are demonstrated according to the job requirement.
2. Select welding job, equipment and job holding devices	2.1 Welding job, equipment and holding devices are identified as per requirement. 2.2 TIG welding machine is prepared as per job requirement. 2.3 Welding equipment and holding devices are set up and adjusted in accordance with the job requirements.
3. Perform GTAW or TIG welding	3.1 TIG welding machine and welding torch performance is checked conforming to the job requirement. 3.2 Amperage and gas flow is set according to the plate thickness and gas flow cup sizes. 3.3 Welding is performed as per the job requirement. 3.4 Welds are cleaned, checked for <u>quality test</u> and <u>defects</u> are identified. 3.5 Defects are rectified to meet the standards of job specifications.
4. Clean/maintain the workplace	4.1 Workplace is inspected to identify any potential hazards. 4.2 Cleaning tasks are performed according to the cleanliness protocols. 4.3 Work areas are organized, and tools and equipment are stored. 4.4 Equipment and machinery are cleaned and maintained. 4.5 The workplace is kept free from debris and obstructions to

	ensure a safe and efficient working environment.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Tools and equipment	1.1 C-clamp 1.2 Chipping hammer 1.3 Ball pin hammer 1.4 Hack saw 1.5 File 1.6 Chisel 1.7 Center punch 1.8 Blacksmith tong 1.9 Scriber 1.10 Pliers 1.11 Wire brush 1.12 Weld gauge 1.13 Welding table with positioner 1.14 Job holding devices/fixture 1.15 Hand grinder 1.16 Hand drill 1.17 GTAW torch 1.18 Electrodes (Tungsten) 1.19 Cups 1.20 Collects 1.21 Gas diffusers 1.22 GTAW power suppliers
2. Base metal/plate	2.1 MS Plates 2.2 Carbon steel 2.3 Stainless steel 2.4 Aluminum
3. Filler metal	3.1 Base metal material 3.2 Stainless steel 3.3 Carbon steel 3.4 Aluminum
4. Electrodes	4.1 Tungsten 4.2 Tungsten alloy
5. Shielding gas	5.1 Inert gas- Argon, helium 5.2 Reactive gases – nitrogen, oxygen or carbon dioxide. 5.3 Mixtures of inert and reactive gases
6. GTAW weld area	6.1 GTAW head 6.2 Power 6.3 Shielding gas 6.4 Contact tube 6.5 Tungsten electrode (non-consumable). 6.6 Weld bead 6.7 Direction of weld

	6.8 Filler rod 6.9 Electric arc
7. Welding joint and position	7.1 Lap joint 7.2 Butt joint 7.3 Tee joint 7.4 1F, 2F and 1G, 2G
8. Quality test	8.1 Destructive test 8.1.1 Tensile test 8.1.2 Bend test 8.1.3 Hardness test 8.1.4 Break test 8.2 Non-Destructive test: 8.2.1 Radiographic test 8.2.2 Ultrasonic test 8.2.3 Magnetic particle test 8.2.4 Leak test 8.3 Visual test
9. Defects	9.1 Lack of penetration 9.2 Excess of penetration 9.3 Porosity 9.4 Crack 9.5 Slag Inclusion 9.6 Distortion 9.7 Undercut 9.8 Lack of fusion 9.9 Burn through 9.10 Spatter 9.11 Notches 9.12 Irregular shape

Curricular Content Guide

1. Underpinning Knowledge	1.1 Process of developing welding plans based on the provided specifications. 1.2 Read and interpret Gas Tungsten Arc Welding (GTAW) or TIG welding drawings. 1.3 Process of selecting base metals, filler metals, tungsten electrodes, and shielding gas. 1.4 Methods for preparing the base metals and the welding area (cleaning, grinding, or smoothing) 1.5 Identifying and selecting the welding job, equipment, and job holding devices based on the specifications. 1.6 The process of setting up and adjusting welding equipment and holding devices. 1.7 Adjust the amperage and gas flow rate based on the plate thickness and the type of gas flow cup used. 1.8 Welding techniques to perform TIG welding on various joints and positions. 1.9 Understand the process of organizing work areas and
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	ensuring tools and equipment.
2. Underpinning Skills	2.1 Selecting the TIG welding machine, tools, and equipment according to the job requirements. 2.2 Preparing base metals and the GTAW welding area to meet the required conditions for welding. 2.3 Preparing the TIG welding machine based on the specific job requirements, ensuring it's ready for use. 2.4 Checking the performance of the TIG welding machine and welding torch. 2.5 Setting the appropriate amperage and gas flow based on the plate thickness and gas flow cup sizes. 2.6 Performing welding according to the job requirements. 2.7 Cleaning the welds and check for quality. 2.8 Cleaning and maintaining equipment and machinery.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 TIG welding machine with accessories 4.5 Materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and prepared work requirements 1.2 selected working job, equipment and job holding devices 1.3 performed GTAW or TIG welding 1.4 cleaned/maintained the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	TIG and MIG WELDING
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom-350 sft (33 sqm)
- Workshop/lab-800 sft (75 sq) OR
- Classroom cum workshop-1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computers/ Laptop	01
2.	Multimedia Projector	01
3.	Internet connectivity	01
4.	AC/DC Arc welding machine	10
5.	Gas welding set	03
6.	MIG welding machine	05
7.	TIG welding machine	05
8.	Gas cutting set	02
9.	Spare cylinder set (Argon, oxygen and acetylene)	01
10.	Power grinding machine	01
11.	Hand grinding machine	04
12.	Bench vice	10
13.	Tool box set for welding workshop	10
14.	Fire extinguisher	04
15.	Welding booth with exhaust system	10
16.	First aid box	03

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on TIG and MIG Welding, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.